Henry Chen

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EDUCATION

University of California, Berkeley | B.A. Statistics; Computing, Data Science & Society — Aug 2026 - May 2030

- **Completed:** CS 61A Structure & Interpretation of Computer Programs; CS 70 Discrete Mathematics & Probability Theory
- **In progress (Fall 2026):** CS 61B Data Structures; DATA C8 Foundations of Data Science; Math 54 Linear Algebra & Differential Equations

TECHNICAL EXPERIENCE & PROJECTS

Tennis IQ | *Project Lead* | 1st Place, Congressional App Challenge (District NV-03) — Oct–Dec 2024

- Developed a mobile application to analyze tennis footage, track joint angles frame by frame, and compare player biomechanics against professional athletes.
- Led a team of four in development using Flutter, Flask, and OpenCV; presented at #HouseOfCode in the U.S. Capitol.
- Designed and implemented a video analysis pipeline using pretrained pose-estimation models and rear-view footage, eliminating the need for wearable sensors or multi-camera setups.

California Institute of Technology / LunaX | *Software Development Intern* | Pasadena, CA — Summers 2024 & 2025

- Developed construction and transportation systems for a Unity-based lunar base simulation under the guidance of Dr. Jonathan Jiang, Principal Scientist at NASA's Jet Propulsion Laboratory.
- Designed and implemented the core tiling and networking infrastructure supporting the simulation's base construction system.
- Developed pathfinding algorithms for optimizing transportation routes between lunar base sites using real lunar terrain data.
- Implemented game-logic refactoring, 3D rendering enhancements, and user-interface improvements.

Polytrack-ML | Reinforcement Learning Project — September 2026 - Present

- Validated a headless Node.js simulator against 27 official leaderboard replays, identifying and correcting physics discrepancies to achieve 100% reproducibility.
- Optimized the simulator's physics loop to process approximately 52,000 frames per second.
- Improved a pure-pursuit controller via parallel parameter search, achieving lap times within 1% of the human record.
- Designed and implemented a PyTorch training pipeline for PPO and behavior cloning, supporting both track-specific and multi-track agents.

LEADERSHIP & TEACHING

DBHS Hackathon Club | *Co-Founder & Co-President* | Diamond Bar, CA — Aug 2024–2026

- Grew a founding team of four into a 70+ member organization and ran a weekly workshop series teaching algorithms and computer science fundamentals to members preparing for competition
- Organized Brahma Hack for 30+ participants; developed programming problems and script-generated test cases.
- Assembled teams for external competitive programming contests and secured faculty sponsorship, a campus STEM-program partnership, and event logistics

Kode4Kids | *President; formerly Treasurer & Scratch Director* | Diamond Bar, CA — Dec 2023–2026

- Led a volunteer organization delivering free online programming instruction in Scratch, JavaScript, Java, and Python, reaching 100+ students and mentoring roughly 15 instructors, and guided the group through nonprofit incorporation
- Authored the current curriculum, including lesson plans, reusable templates, and quiz games, and personally taught 50+ students, many of whom returned for more advanced tracks
- Ran fundraising as treasurer and built the Scratch program as its director, recruiting instructors and coordinating enrollment

SKILLS & AWARDS

Languages: Python, Java, C++, SQL, JavaScript, Dart, C#.
Data & ML: PyTorch, pandas, scikit-learn, OpenCV; reinforcement learning (PPO), behavior cloning, pose estimation, supervised and unsupervised modeling.
Frameworks & Tools: Node.js, Flask, Django, React, Tailwind CSS, Flutter, Unity, Git/GitHub, Google Cloud, Linux, SQLite, MySQL
Awards: USACO Silver Division; 1st Place, Congressional App Challenge (District NV-03); 5th Place in California, FBLA Website Development (SLC 2025).